Research Experience

Rachel N. Slaybaugh

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DCVC Palo Alto, CA

Partner *Feb. 2023 – present*

Principal *Feb. 2022 – Feb. 2023*

- Source, select, fund, and support climate tech companies
- Independent board member: Radiant Nuclear (04/2022 - present)
- Board observer: Brimstone (02/2022 - present), Petra (08/2022 - 06/2024), and Verdigris (10/2023 - present)
- Board member: Fervo Energy (08/2022 - present) and People and Culture Committee Chair, Fourth Power (10/2023 - present), Equilibrium Energy (12/2023 - present)

Lawrence Berkeley National Laboratory Jan. 2021 – Dec. 2021

Cyclotron Road Division Director *Berkeley, CA*

- Source and select ∼10 hard tech innovators per year for a 2-year fellowship program to turn their technology concept into a product with a positive societal impact (∼$6M/year)
- Fellows embed at LBNL where we support their technical development and collaborations
- Responsible for outcomes, safety, and reporting to Department of Energy and LBNL leadership

University of California, Berkeley Berkeley, CA

Associate Professor of Nuclear Engineering *July 2020 – Dec. 2021*

Assistant Professor *Jan. 2014 – July 2020*

- Founded the Nuclear Innovation Bootcamp, which brings diverse students from around the world to learn skills essential to innovation in nuclear energy
- Developing numerical methods for neutral particle transport with an emphasis on supercomputing and advanced architectures; applications in reactor design, shielding, and nonproliferation
- Also mentored PhDs in optimization, thermal fluids, and cryptography and anomaly detection
- Published 25 journal articles, 44 refereed conference proceedings, 3 technical reports, 2 book chapters, 5 open source pieces of software, and 2 policy pieces
- Graduated 9 PhD and 4 MS students; research adviser for 1 assistant project scientist, 1 postdoctoral scholar, 1 visiting scholar, and 15 undergraduate students
- Won >$2.5M as principal investigator (PI) and >$26M as co-PI for numerical methods research
- Created and support a course in which Berkeley students do hands on science experiments at underserved elementary schools in Oakland

Advanced Research Projects Agency – Energy Jan. 2017 – Oct. 2020

Program Director *Washington, DC*

- Created programs and selected teams developing technologies for advanced nuclear fission reactors, $95M across 28 teams: MEITNER, the Nuclear OPEN+ cohort, LISE, and GEMINA
- Provided deep technical and business guidance while managing: the nuclear teams; the TERRA and ROOTS Programs, supporting research for sensing and data analytics for above- and below-ground plant outcomes (∼$90M across 18 teams); and the FOCUS Program, supporting research for solar technologies that combine photovoltaic and concentrated solar power technologies (∼$12M across 4 teams)

- Supported program creation in other energy areas through workshops, brainstorming, and feedback

Bettis Laboratory West Mifflin, PA
Senior Engineer in the Shield Design and Development group *Mar. 2012 – Aug. 2014*

- Implemented the Forward-Weighted Consistent Adjoint Driven Importance Sampling (FW-CADIS) method and developed new resonance factor for variance reduction in Monte Carlo
- Qualified these methods and software for use in shield design to dramatically reduce time and improve accuracy in design calculations
- Scientific Software Development Committee: leader in developing internal website for sharing software carpentry tools and resources

University of Wisconsin–Madison Madison, WI
Research Assistant / Rickover Fellow *Sept. 2006 – Nov. 2011*

- Dissertation: "Acceleration Methods for Massively Parallel Deterministic Transport" where I added 3 new methods to Denovo, software from Oak Ridge National Laboratory, that are still used
- Developed two Monte Carlo source sampling methods for arbitrarily shaped plasma sources

Penn State Breazeale Reactor University Park, PA
Reactor Operator *Aug. 2003 – Apr. 2006*

- NRC licensed Reactor Operator for TRIGA Mark III reactor
- Analyzed core burn-up anomaly; calibrated gamma irradiation facilities